ZentroMart AI Voice Assistant

VoiceOps 2026 Capstone Report · Consulting & Analytics Club, IIT Guwahati × Gnani.ai

1. Executive Summary & Architecture

The ZentroMart AI Voice Assistant is a production-grade multimodal customer support solution designed for a fictional Indian e-commerce company handling groceries and electronics. The system delivers policy-grounded factual responses across chat and voice, enforces strict caller identity verification, resolves follow-up context across multi-turn sessions, executes official database lookups and ticketing actions, and uses Gnani's speech models for listening and speaking in 10 Indian languages. In addition, the web interface replicates the minimalist aesthetic of *Cinematic Sites in Ten Minutes*, featuring an ultra-smooth looping background video, floating liquid-glass navigation, and Instrument Serif typography.

End-to-End System Architecture

Component	Implementation Technology	Core Function & Compliance
Ears (STT)	Gnani STT (gnani-vachana SDK)	Transcribes 10 Indian languages; en-IN Hinglish support; <=60s clips
Brain (Agent)	Google Gemini + Custom FSM State Manager	Identity-gated order lookup; coreference resolution; policy decisions
Knowledge (RAG)	Hybrid TF-IDF & Dense Semantic Vector Store	21 Markdown policies + 70 SKU catalog rows; precedence ranking
Tools Layer	Official synthetic lookup_order & create_ticket	Identity check via phone last 4 / pincode; summary verification
Voice (TTS)	Gnani TTS (timbre-v2.5, Kaveri voice)	Converts grounded answers into natural spoken audio replies
Interface	React 18 + Vite + Tailwind + Framer Motion	Cinematic liquid-glass hero + interactive voice workspace console

2. Step 1: LLM Prompting Baseline

Implementation: Step 1 establishes the baseline prompting agent without access to ZentroMart's private knowledge base or tools. The agent identifies itself as ZentroMart's AI assistant in its opening turn, maintains a polite, spoken-friendly persona, asks clarifying questions when details are omitted, and adheres to strict anti-hallucination boundaries—never fabricating refund dates, amounts, or policy rules.

Observations: While the baseline agent excels at conversational tone and conversational politeness, it lacks domain-specific knowledge. It cannot know that ZentroMart Wallet refunds are instant (within 2 hours), or that fashion products have a 15-day calendar return window, or whether ELE-0012 carries Assured warranty. Consequently, it achieves high decision etiquette but near-zero task accuracy on factual policy questions.

3. Step 2: Document-Grounded RAG Pipeline

Knowledge Base Ingestion: All 21 official Markdown documents in kb/ were parsed and chunked by section header (e.g. ## [RET-2] Return windows by category), preserving frontmatter metadata (doc_id, doc_type, status, effective_from). In addition, all 70 product rows from products.csv were indexed as catalog chunks (products.csv:SKU) to ground warranty claims, return categories, and pricing.

Strict Policy Precedence: In accordance with kb/00_about_this_knowledge_base.md, retrieval strictly enforces document hierarchy: **Current Policy (Rank 1) > Agent Guides (Rank 2) > Help Articles (Rank 3) > FAQs (Rank 4) > Marketing (Rank 5)**. Crucially, pol_returns_v4.md takes precedence over pol_returns_v3_superseded.md, and policy REF-2 overrides the trap in FAQ-1 (which incorrectly claims 5 to 7 days for all refunds).

Date Rules: All timelines are pinned to the capstone reference date: **20 October 2026**. Business days run Monday to Saturday, skipping the 4 warehouse closure dates in holidays.csv (15 Aug, 02 Oct, 19 Oct, 09 Nov 2026). Return windows use calendar days. Dates are spoken with the full year.

Step 1 vs. Step 2 Practice Evaluation Benchmark (52 Questions)

Evaluation Metric	Step 1: Baseline	Step 2: Grounded RAG	Improvement / Impact
Decision Accuracy (answer/clarify)	80.77%	**88.46%**	+7.69% higher decision precision
Key Facts Rate (must_include)	7.69%	**25.0%**	3.25× increase in factual correctness
Hallucination Avoidance (must_not_include)	100.0%	**100.0%**	Zero false policy promises or fake dates
Evidence Retrieval Hit Rate (gold_evidence)	0.0% (N/A)	**75.0%**	Correct section retrieved in top-3
Overall Pass Rate (Strict)	1.92%	**19.23%**	**10.0× overall improvement**

Abstention & Unsupported Inquiries

For queries completely absent from company documents (e.g. PQ0004 midnight delivery slots, PQ0009 student discounts, custom gift wrapping), the agent abstains cleanly: *"Sorry, I don't have information about that. I can connect you to a colleague who can help."* Under no circumstances does the system fabricate policies.

4. Step 3: Multi-turn Agent & Tool Integration

Session State & Coreference: The agent maintains conversational state across turns, remembering the customer identity, active order_id, verification status, and previous tool outputs. When customers ask elliptical follow-ups like *"How long will that take?"* or *"Can I cancel it now?"*, the agent resolves context to the active order rather than demanding redundant repetitions.

Identity Gating & Privacy Enforcement: The agent integrates the official lookup_order(order_id) tool. When an order number is introduced, the tool returns verification_required=True with zero order disclosures. The agent asks the customer to provide either their 4-digit registered phone suffix or 6-digit delivery pincode. Order status, line items, and addresses are revealed **only after** lookup_order(order_id, verification_value) returns verified=True. Max one retry is permitted; details are never leaked.

Human Escalation & Ticket Logging: When policy requires human intervention (e.g. damage claims on high-value electronics exceeding ■10,000, or unauthorized compensation demands), the agent invokes create_ticket(summary, order_id, reason, intent). The agent confirms handover **only if** the tool responds with created=True and communicates the ticket ID.

Multi-turn Evaluation Benchmark (23 Scenarios, 96 Turns)

Conversation Benchmark Criterion	Measured Score	Target / Standard	Evaluation Status
Caller Verification Gating Compliance	**100.0%**	100% (No leaks)	**FLAWLESS PASS**
Tool Invocation Accuracy (lookup/ticket)	**87.04%**	>= 80%	**EXCEEDS TARGET**
Turn-by-Turn Decision & Phrasing Pass	**75.0%**	>= 70%	**HIGH SUCCESS**

Representative Turn Trajectory (Scenario PC002)

- **Turn 1 (Intro):** Customer greets; agent introduces itself as AI assistant and requests Order ID (STY-1).
- **Turn 2 (Gating):** Customer provides '8978 2837'; agent calls lookup_order(89782837) and requests phone suffix / pincode (IDV-2).
- **Turn 3 (Verification):** Customer provides '4533'; agent verifies caller Kabir and discloses order status.
- **Turn 4 (Escalation):** Customer requests instant replacement on ■34,999 Smart TV; agent refuses direct approval and creates ticket TKTa807e788b5284... (DWM-3, ESC-1).

5. Step 4: Gnani Speech Integration & Batch Runner

Native Gnani Pipeline: Speech processing is powered exclusively by the official gnani-vachana SDK: GnaniSTTClient for speech-to-text and GnaniTTSClient for text-to-speech. Per the strict competition rules, **no third-party STT/TTS fallbacks** (such as Whisper, Web Speech, or ElevenLabs) are utilized. When GNANI_API_KEY is configured, the browser microphone and audio file uploads stream audio to Gnani STT, process the transcript through the Step 3 agent, and synthesize spoken answers via Gnani TTS (timbre-v2.5, Kaveri voice).

Batch Audio CLI Command: As required by the evaluation protocol, the standalone runner executes with:

```
python run.py --input-dir ./input_audio --output-dir ./outputs
```

The batch runner discovers all audio clips deterministically, passes them through STT, Agent, and TTS, and produces manifest.json, results.csv, generated WAV replies in outputs/audio/, transcripts in outputs/transcripts/, and evidence logs in outputs/evidence/.

6. Engineering Challenges & Lessons Learned

1. **Free-Tier LLM Rate Limits (429):** Rapid sequential evaluation triggers the Gemini Free Tier 5 requests/minute quota. *Solution:* Implemented SHA-256 prompt response caching alongside an evidence-derived fallback engine. This enables immediate, policy-grounded replies even during external network throttling.
2. **Verification Proof String Extraction:** When customers stated *"My phone last 4 digits are 4533"*, raw digit extraction extracted '44533' (treating the word '4' as part of the proof). *Solution:* Implemented word-boundary token matching (\b\d{4}\b) and stripped the order ID prior to proof extraction.
3. **Document Precedence Traps:** FAQ-1 claims 5–7 days for all refunds, contradicting the wallet policy. *Solution:* Strict precedence re-ranking guarantees that current policies (REF-2) always override FAQs.

7. Bonus Work & UI/UX Excellence

• **Cinematic Sites in Ten Minutes UI:** Full-screen 5s looping cinematic background video (hero-loop.mp4), Instrument Serif display typography, floating liquid-glass navigation with 1.4px gradient pseudo-borders, and subtle black/white aesthetics.
• **Live Interactive Grounded Evidence Drawer:** Clickable citation pills allow customers and judges to inspect the exact official clauses used.
• **Multi-Language Support:** Full support across 10 Indian regional languages (Hindi, Hinglish, Tamil, Telugu, Bengali, Kannada, Gujarati, Marathi, Punjabi, Malayalam).